INCH – POUND

MIL-DTL-3918B
4 February 2011
SUPERSEDING
MIL-L-3918A
28 September 1970

MILITARY SPECIFICATION

LUBRICATING OIL, INSTRUMENT, JEWEL BEARING

Inactive for new design after 29 March 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of low-temperature, non-spreading lubricating oil for use on instrumental jewel bearings.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Standards. The following standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DLA Aviation - VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil.

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FEDERAL STANDARD

FED-STD-791	-	Testing Method of Lubricants, Liquid Fuels, and Related Products

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-290	-	Packaging, Packing and Marking of Petroleum and Related Products

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D94	-	Standard Test Methods for Saponification Number of Petroleum Products
ASTM D130	-	Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (the Calculation of Dynamic Viscosity)
ASTM D972	-	Standard Test Method for Evaporation Loss of Lubricating Greases and Oils
ASTM D974	-	Standard Test Method for Acid and Base Number by Color-Indicator Titration
ASTM D1218	-	Standard Test Method for Refractive Index and Refractive Dispersion of Hydrocarbon Liquids
ASTM D1519	-	Standard Test Methods for Rubber Chemicals – Determination of Melting Range
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D4177	-	Standard Practice for Automatic Sampling of Petroleum and Petroleum Products

(Copies of these documents are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

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AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4 - Sampling Procedures and Tables by Inspection of Attributes

(Copies of ANSI/ASQ Z1.4 are available online at www.asq.org or ASQ, 600 North Plankinton Avenue, Milwaukee, Wisconsin 53203)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the oil shall conform to the requirements shown in Table I.

TABLE I. Chemical Composition.

Ingredient	Weight Percent
Benzyl phenylundecanoate (Benzyl p)	59.95 ± 0.50
Diethylene glycol di-n-caproate (D glycol)	39.45 ± 0.50
Dodecylpiperidine stearate (D stearate)	0.40 ± 0.10
p-tertiary - butyl catechol (Butyl c)	0.20 ± 0.10

3.1.1. Ingredient properties. The ingredients shall have properties in accordance with table II when tested in accordance with 4.4.1.

TABLE II. Ingredient property requirements.

Ingredient	Viscosity at 100°F (38°C) (mm^2/s)	Refractive Index	Acid No.	Saponification No.	Melting Point
Benzyl p	15.7 – 16.7	1.517 – 1.520	< 0.10	159±1	---
D glycol	5.0 – 5.5	1.437 – 1.440	<0.10	---	---
D stearate	---	---	105 - 115	---	Initial - 97±4°F (36±2°C) Final - 102±4°F (39±2°C)
Butyl c	---	---	---	---	133°F - 135°F (56°C - 57°C)

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3.2 Oil properties.

3.2.1 Viscosity. The viscosity of the oil, when determined at 100°F (38°C) as specified in 4.4, shall not be less than 9.5 millimeter^2 per second (mm^2/s) nor more than 10.5 mm^2/s.

3.2.2 Acid number. The total acid number of the oil, when determined as specified in 4.4, shall not exceed 0.60.

3.2.3 Evaporation loss. The evaporation loss of the oil, when determined as a temperature of 210°F (99°C) in accordance with 4.4, shall not exceed 2.0 percent.

3.2.4 Copper strip corrosion. When the oil is tested for 3 hours at 212°F (100°C) as specified in 4.4, the discoloration of the copper strip shall not be greater than 1a rating.

3.2.5 Corrosion and oxidation stability. When the oil is tested as specified in 4.4 it shall meet the following requirements:

3.2.5.1 Corrosion. The change in weight of copper and steel specimens shall not exceed 0.5 milligrams per square centimeter of surface. There shall be no pitting, etching, or visible corrosion of the metal surfaces. A slight coloration of the metal surfaces shall not be objectionable.

3.2.5.2 Oxidation. After completion of the corrosion oxidation test, the viscosity of the oil shall not have changed more than ±5 percent of the original viscosity, the change in acid number of the oil shall not exceed 0.15, and the volatile acids evolved into the delivery tube shall not require more than 0.5 milligrams of KOH, per gram of oil, for neutralization.

3.2.6 Spreading on metal. When the oil is tested as specified in 4.4.2.1, the average diameter of three drops of oil on polished steel shall not increase by more than 5 percent. The residue from the corrosion test (see 3.2.5.2) shall not spread by more than 5 percent in 168 hours (7 days).

3.2.7 Low temperature stability. After a dry sample of the oil has been tested as specified in 4.4 for 48 hours at -40°F (-40°C), the oil shall not crystallize, separate or show evidence of precipitates. Cloudiness in the oil shall not be considered objectionable.

3. 2.8 Workmanship. The oil shall contain no perfume or dye and shall be visibly clear and free of sediment.

4. VERIFICATION

4.1 Classifications of inspection. The inspection requirements specified herein are all conformances.

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4.2 Sampling.

4.2.1 Sampling for examination of the preparation for delivery.

4.2.1.1 Sampling for inspection of containers. A random sample of packed containers from each lot shall be taken in accordance with ANSI Z1.4 at inspection level II. The acceptance criteria shall be as specified in the procurement document (see 6.2).

4.2.1.2 Sampling for tests. The samples for tests shall be eight ounces of oil taken from the bulk lot in accordance with ASTM D4057 or ASTM D4177 and a sample of each of the ingredients from each lot of oil offered or processed for delivery under a contract or order. The acceptance criteria shall be as specified in the procurement document (see 6.2).

4.3 Examination of the preparation for delivery. Samples taken in accordance with 4.2.1 shall be examined for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, and packing and marking requirements.

4.4 Inspection methods.

4.4.1 Chemical composition. The contractor shall certify that the composition of the oil is in accordance with table I. The ingredients shall be tested in accordance with table II as required.

Table III. Test methods for ingredient properties.

Method Title	ASTM Method No.
Viscosity	D445
Refractive Index	D1218
Acid Number	D974
Saponification Number	D94
Melting Point	D1519

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4.4.2. Oil test methods. The tests listed in Table IV shall be performed in accordance with the applicable method.

TABLE IV. Test methods for oil properties.

Requirement Paragraph	Method Title	FED-STD-791 Method No.	ASTM Method No.	Paragraph herein
3.2	Viscosity		D445	
3.3	Acid Number		D974	
3.4	Evaporation Loss		D972	
3.5	Copper Corrosion by Petroleum Products		D130	
3.6	Corrosiveness and Oxidation Stability of Light Oils (Metal Strip)	5308		
3.7	Spreading of Oil			4.4.2.1
3.8	Low Temperature Stability	3459		
3.9	Workmanship			4.4.2.2

4.4.2.1 Spreading on metal.

4.4.2.1.1 Preparation of metal surface. A steel surface with a minimum area of 0.25 square inches shall be polished with metallographic papers Nos. 0, 2/0, 3/0, and 4/0 in that order and finally with Fisher's "Gamal" or equivalent polishing alumina on a wheel covered with heavy broadcloth. After the surface has been polished, it shall be washed with water, swabbed with wet absorbent cotton, and rinsed thoroughly with distilled water. The surface shall then be rinsed with reagent grade 99 percent methanol and dried for 15 minutes in an oven at 221°F (105°C). The specimen shall then be allowed to cool to room temperature in a desiccator.

4.4.2.1.2 Procedure. Three drops of oil shall be placed on the metal surface with a clean platinum wire. The drops shall be between 1.0 and 2.0 mm in diameter. (A platinum wire 0.005 inches in diameter is suitable for depositing a drop this size.) After the wire has been recleaned, two drops of reagent grade orthotricresyl phosphate shall be placed on the metal surface in a similar manner. The metal and the drops shall be covered to keep out dust and allowed to stand for 15 minutes. The diameters of each of the five drops shall be measured with a measuring apparatus having an accuracy of one percent and the values recorded. The metal shall be carefully transferred to an air bath maintained at 113 ± 3°F (45 ± 2°C) and allowed to remain for 24 hours. The drops shall be measured. If both the oil and the orthotricresyl phosphate appear to be spreading, the test should be repeated on a freshly prepared surface. If no evidence of spreading is observed the test is continued for an additional 144 hours (6 days). After the 168 hours (7days), the diameter of the oil drops is measured and any increase in diameter calculated.

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4.4.2.1.3 Results. The increase in diameter of each drop shall be calculated as follows:

$$\text{Percent increase} = \frac{D_7 - D_0}{D_0} \times 100 \text{ when}$$

D_7 = diameter of drop after 7 days

D_0 = diameter of drop after 15 minutes

The average of the values for three drops shall be reported as the percent spreading in a week.

4.4.2.2 Workmanship. The oil shall be inspected for compliance with the workmanship requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The oil covered by this specification is intended for the lubrication of steel pivot and jewel bearing combinations in time pieces or other fine instruments. It does not spread on highly polished, clean metal or jewel surfaces but may tend to spread when used on unpolished surfaces, especially when it is contaminated with dust or other foreign materials. It will allow operation of most instrument mechanisms at temperatures as low as -40°F (-40°C).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Acceptance criteria (see 4.2.1.1)
c. Quantity of oil required
d. Packaging requirements (see 5.1).

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6.3 Definitions.

6.3.1 Bulk lot. An indefinite quantity of a homogenous mixture of material offered for acceptance in a single isolated container; or manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.3.2 Packaged lot. An indefinite number of unit containers of identical size and type, offered for acceptance, and filled with a homogenous mixture of material from one isolated container; or filled with a homogenous mixture of material manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.4 Subject term (key word) listing.

Benzyl phenylundecanoate
Diethylene glycol di-n-caproate
Dodecylpiperidine stearate
Methanol
Orthotricresyl phosphate
p-tertiary-butyl catechol

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AR
Navy – AS
Air Force – 11
DLA – GS

Review Activities
Army – AT, AV, MI
Navy – OS, SA, SH
Air Force – 68

Preparing Activity:
DLA – GS3

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